

The background features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

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SAS408

Chaos and SaaS: Pushing your multi-tenant architecture

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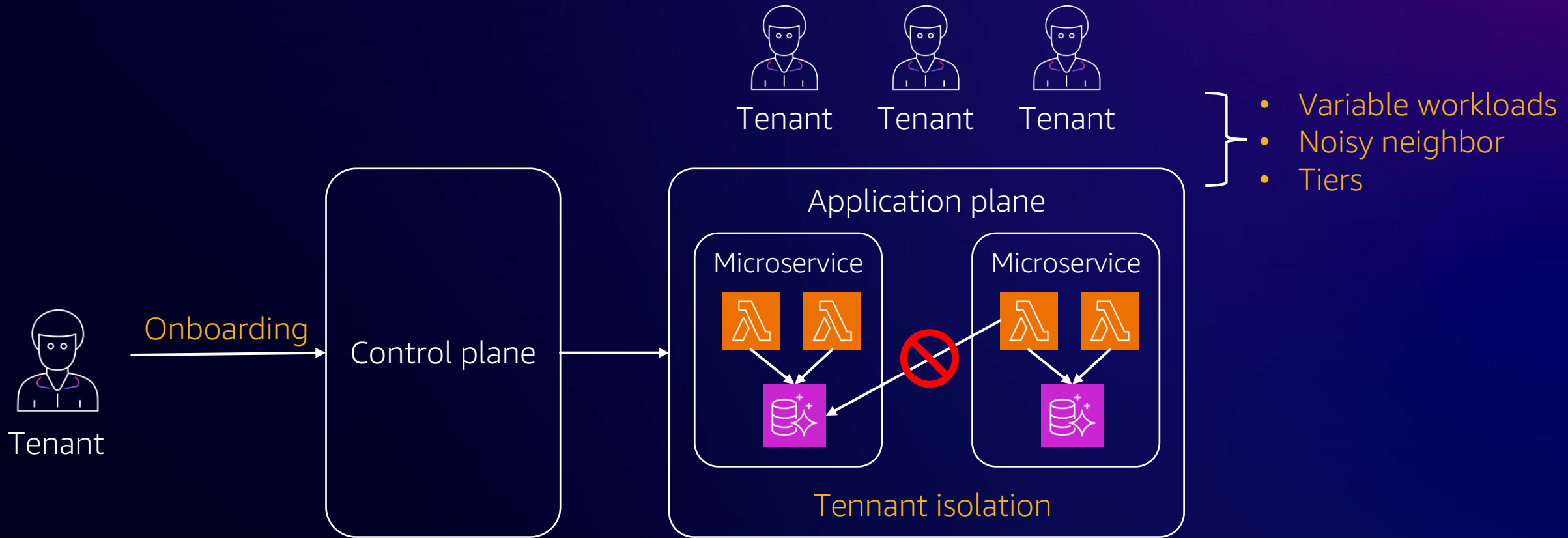
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Senior Solutions Architect
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Tenancy and chaos: A new wrinkle



- What are the failure patterns and stress point in multi-tenant environments?
- Where do we need to introduce/simulate chaos?

Why it matters



How can we support new tenant growth?



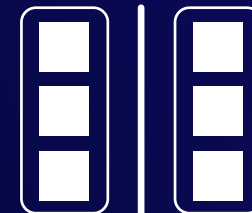
How can we be sure we'll scale efficiently?



How do we know if different tenant workloads will tip the system over?

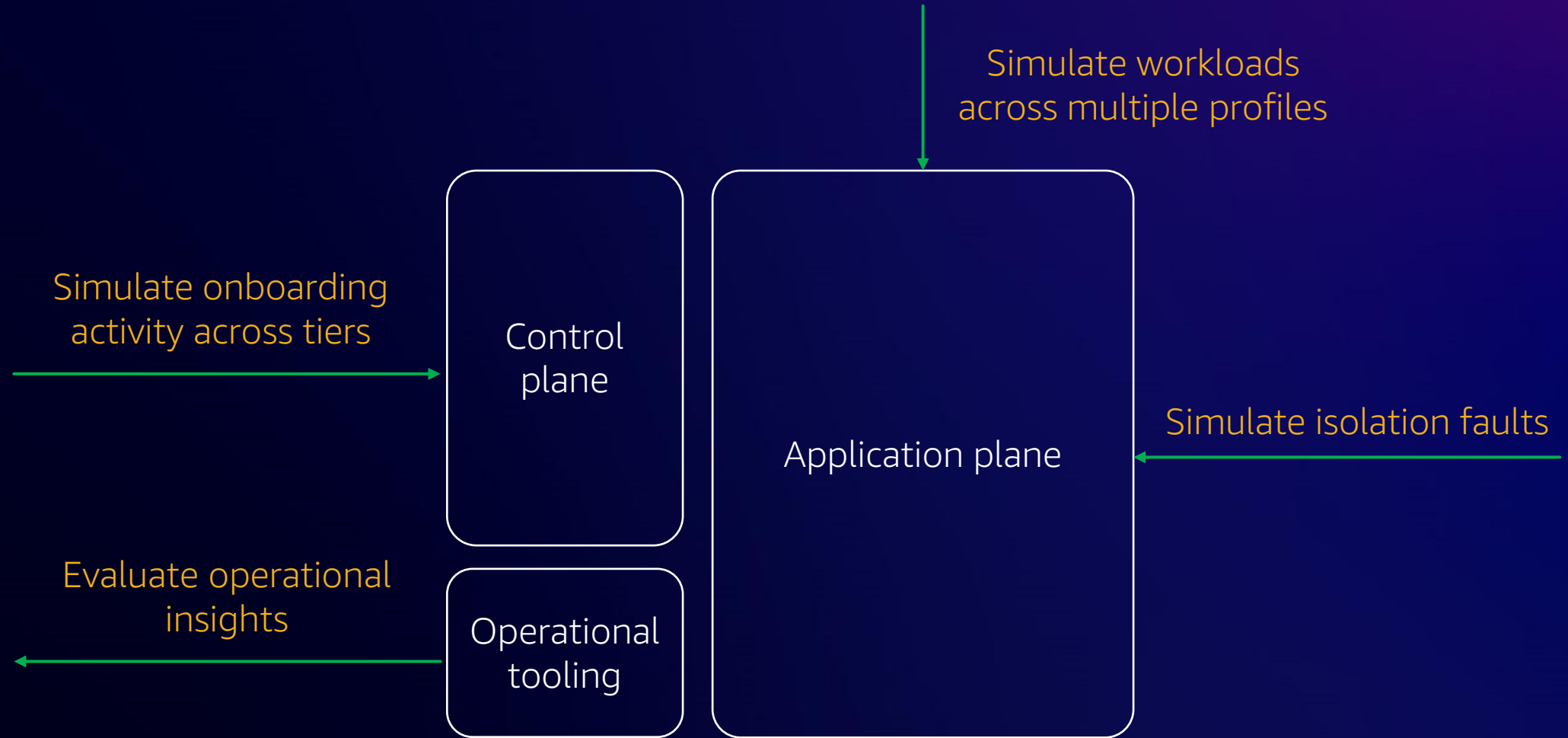


How do we know we'll meet tenant SLAs?

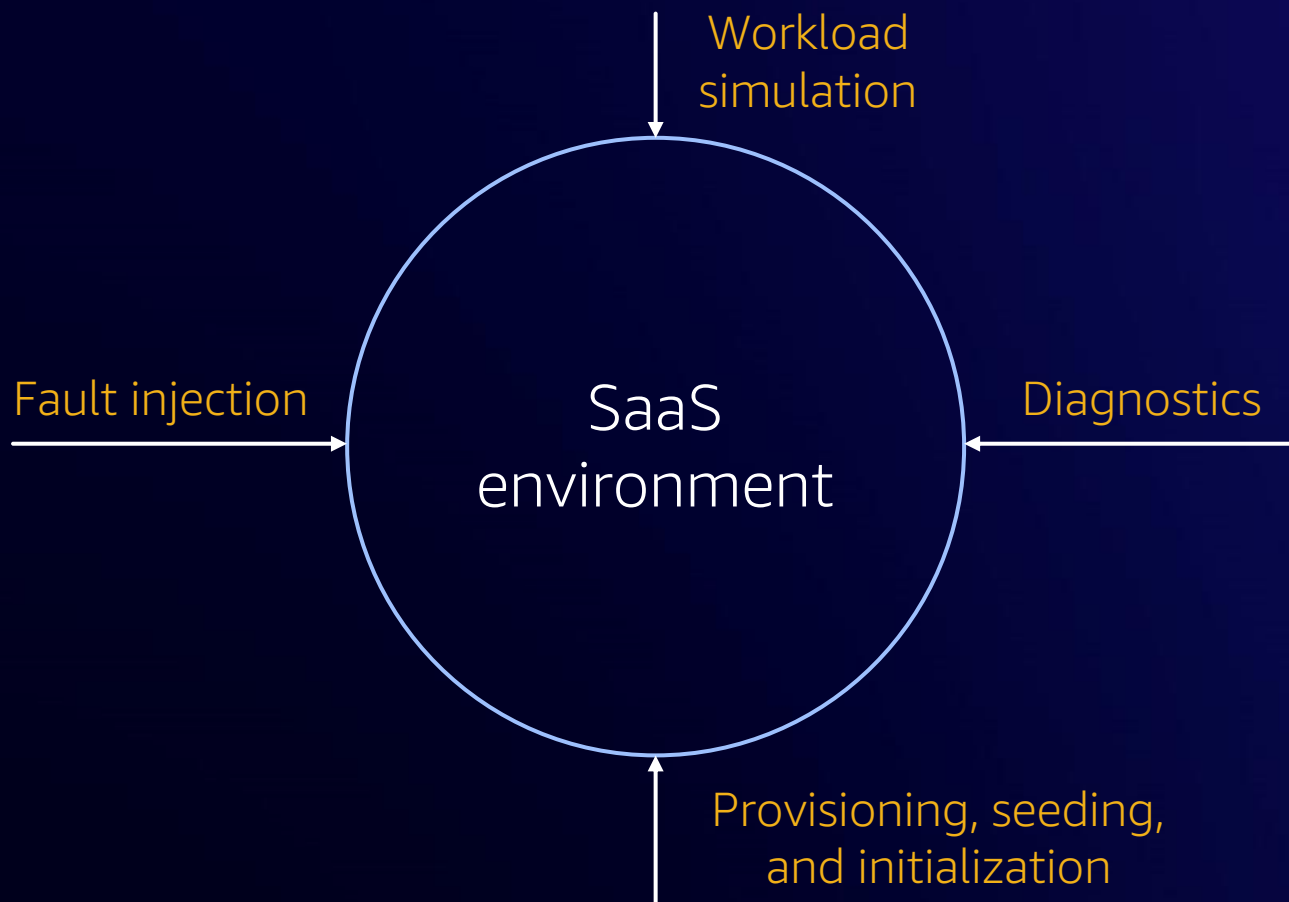


How do we know isolation works?

The hard part



The tooling landscape



- **Simulation**
 - Build your own
 - 3P libraries/tools
- **Diagnostics**
 - CloudWatch
 - OpenSearch
 - Partner observability tools
- **Fault injection**
 - Stack dependent
 - Sidecars
 - Wrappers
 - Fault inject libraries/services

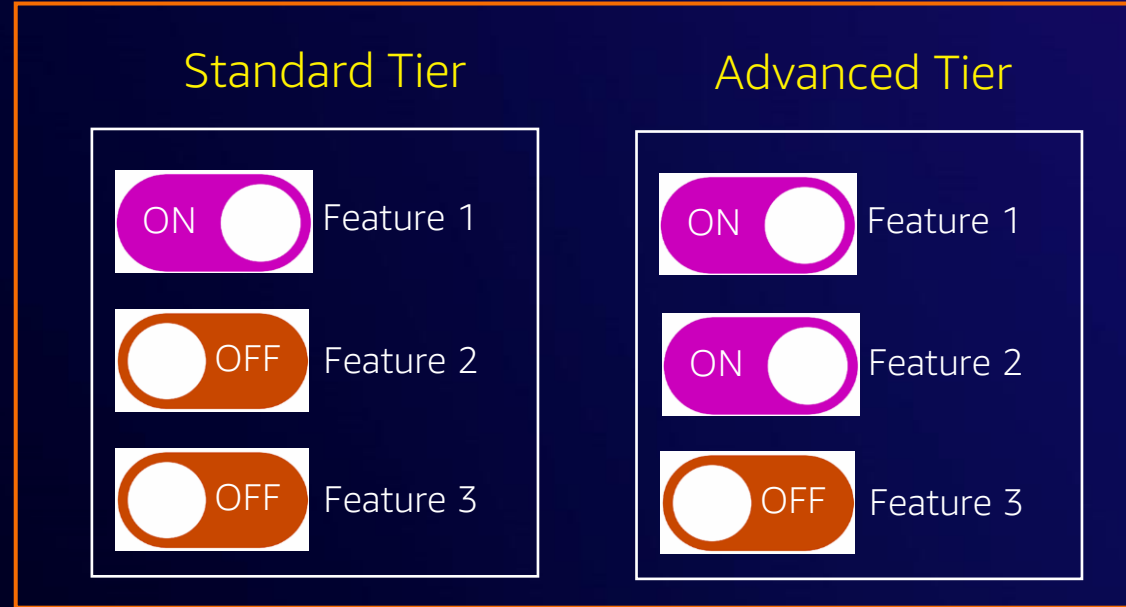
Developing a workload profile

Defining your chaos profile

Tenant personas



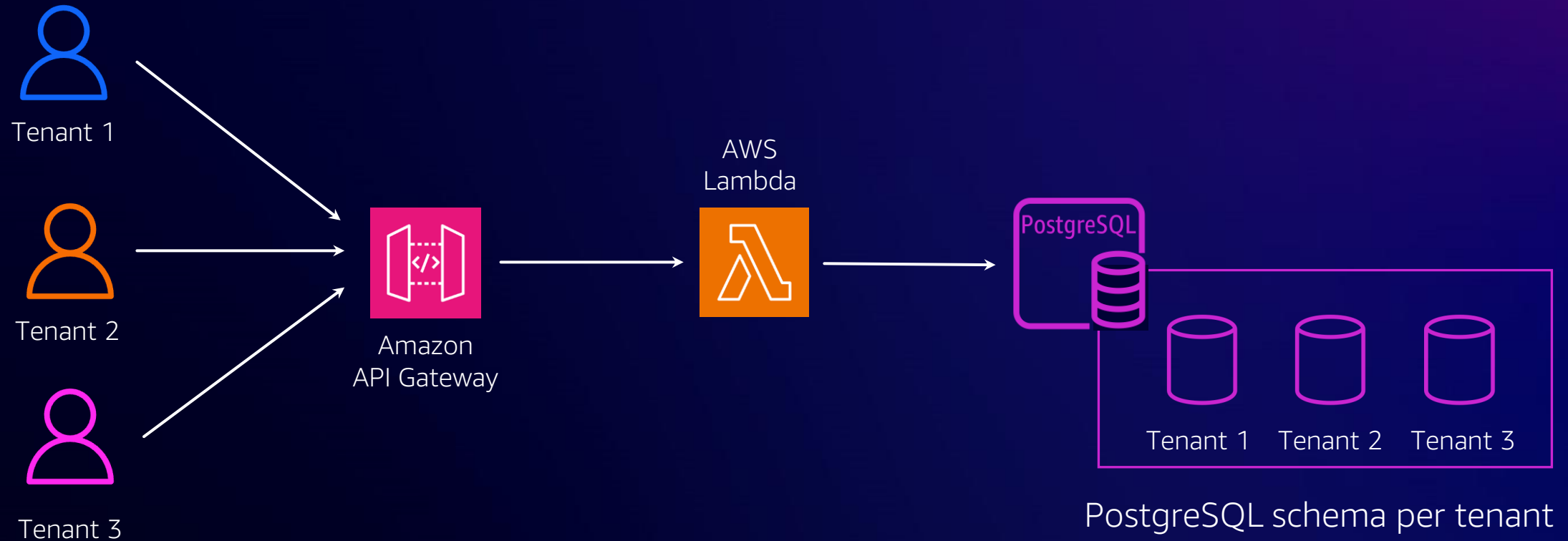
Tiering essential to workload profiling



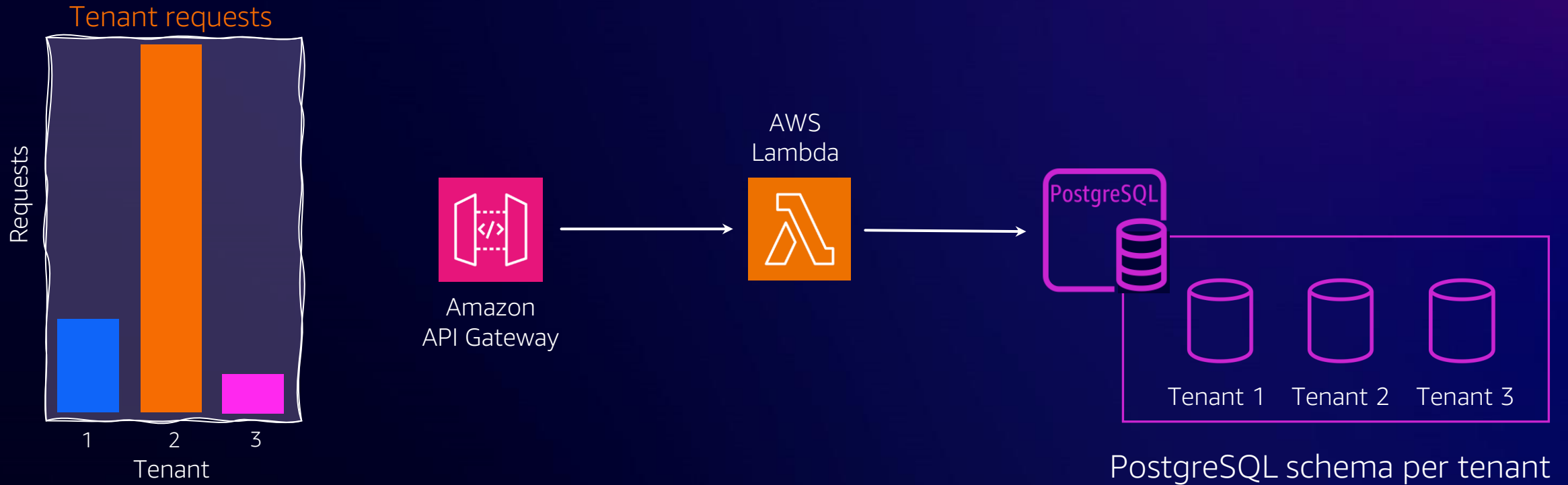
- Features are distributed across tiers
- What effect does this have on workload simulation?

Chaos and noisy neighbor

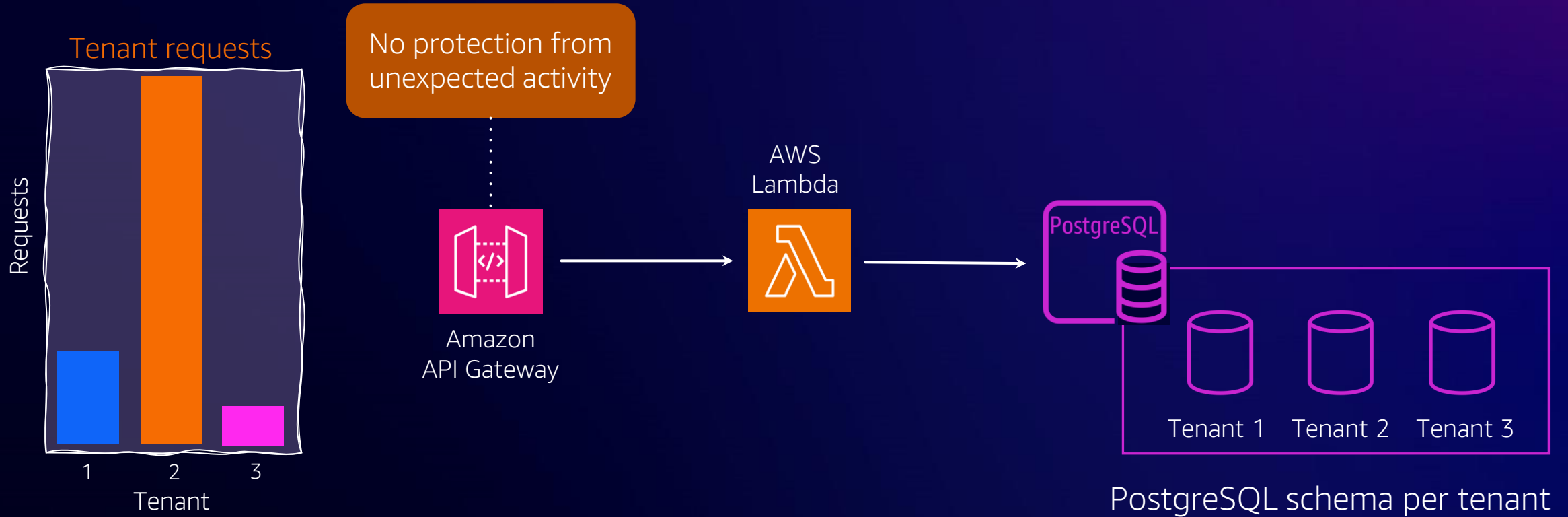
What is noisy neighbor and where can it be in SaaS?



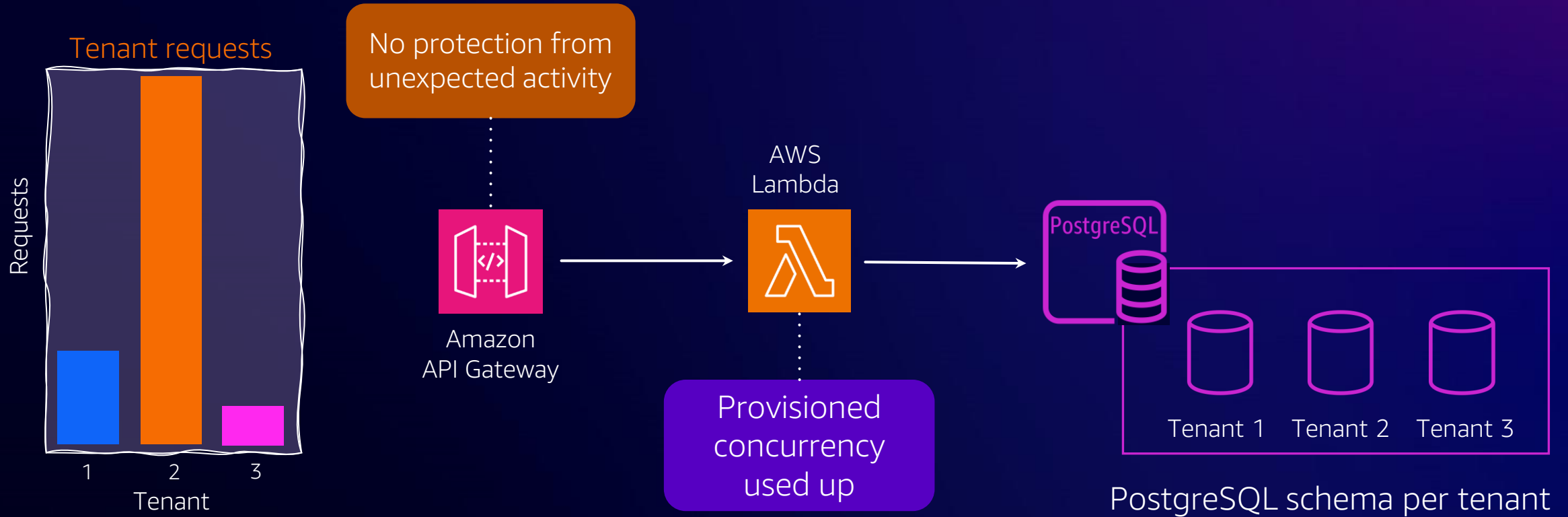
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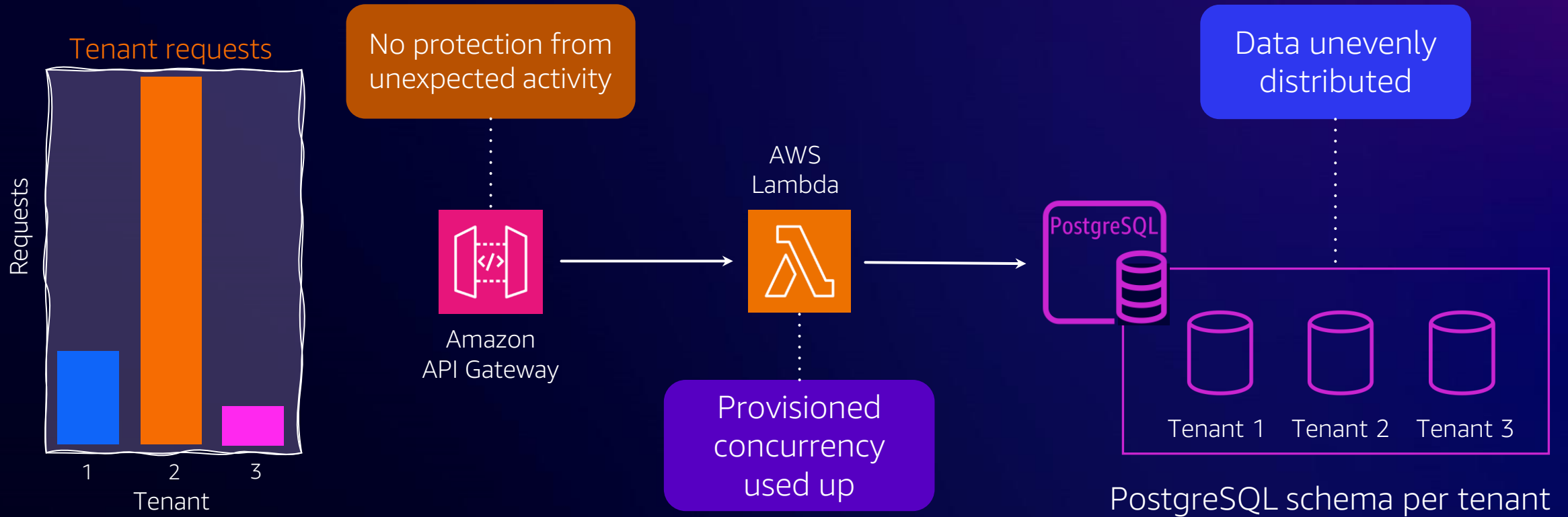
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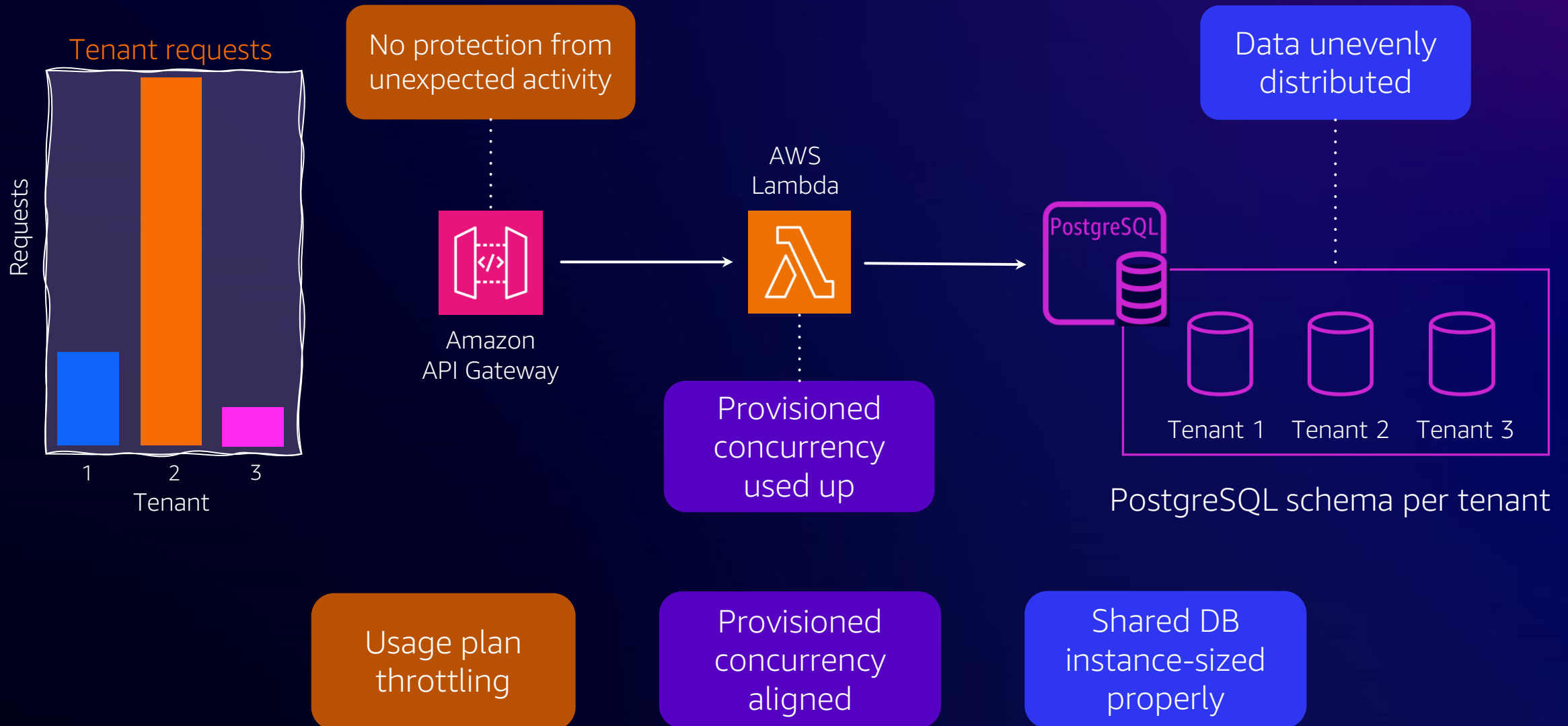
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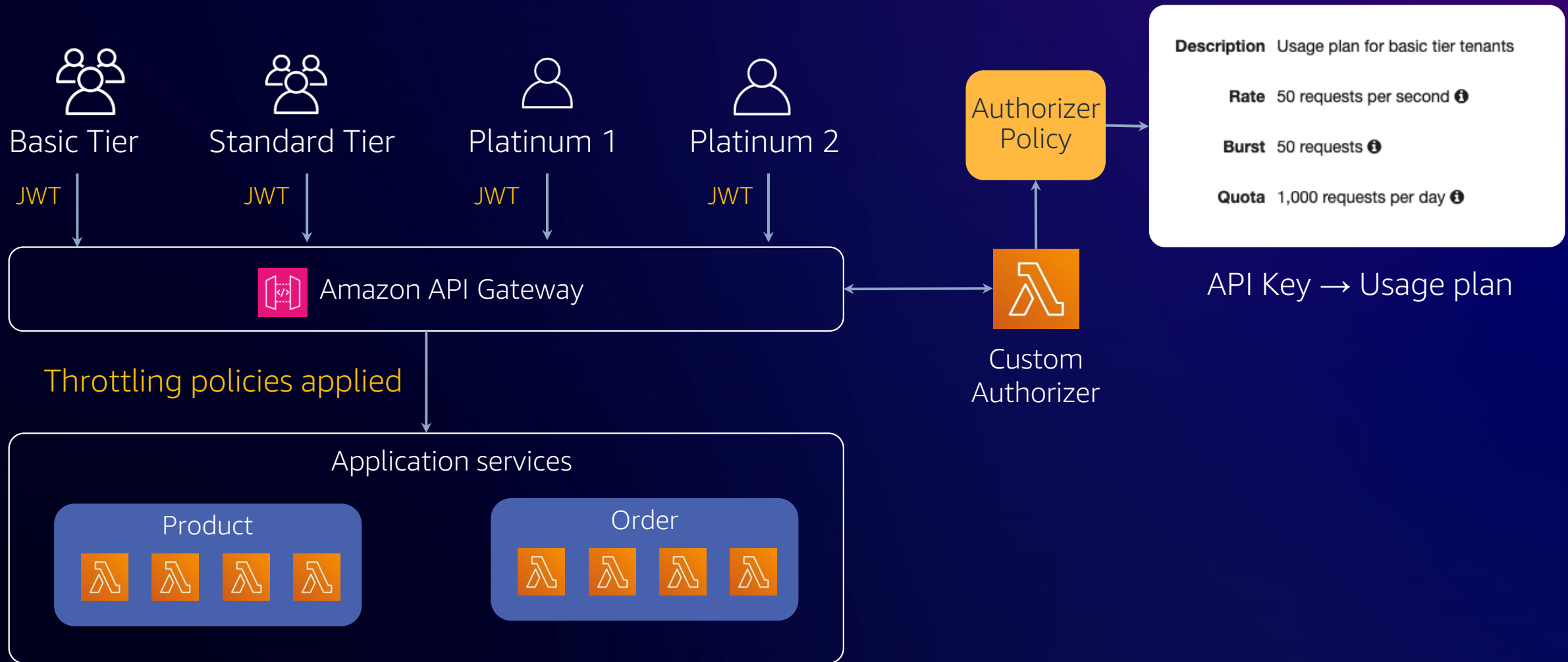
What is noisy neighbor and where can it be in SaaS?



What is noisy neighbor and where can it be in SaaS?



It starts with having a noisy neighbor strategy



Making authorizer the source

```
600     DefinitionBody:
601         openapi: 3.0.1
602         ...
603         x-amazon-apigateway-api-key-source : "AUTHORIZER"
```

```
116     #pass sts credentials to lambda
117     context = {
118         'accesskey': credentials['AccessKeyId'], # $context.authorizer.key -> value
119         'secretkey' : credentials['SecretAccessKey'],
120         'sessiontoken' : credentials["SessionToken"],
121         'userName': user_name,
122         'tenantId': tenant_id,
123         'userPoolId': userpool_id,
124         'apiKey': api_key,
125         'userRole': user_role
126     }
127
128     authResponse['context'] = context
129     authResponse['usageIdentifierKey'] = api_key
130
131     return authResponse
```

Exercising your throttling policies



Noisy neighbor
(scenario 1)



Noisy neighbor
(scenario 2)



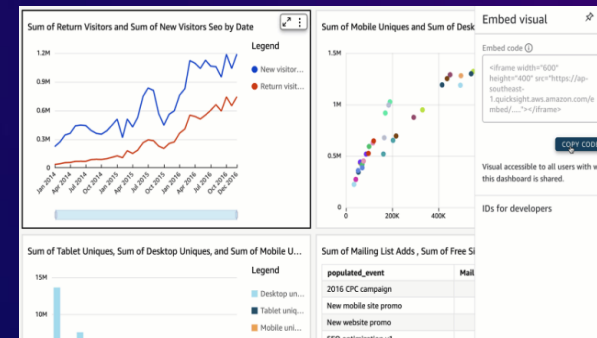
Noisy neighbor
(scenario N)

Single basic tier
noisy neighbor

Two advanced tier
noisy neighbors

Multiple basic tier
noisy neighbors

Observe
throttling policies



1

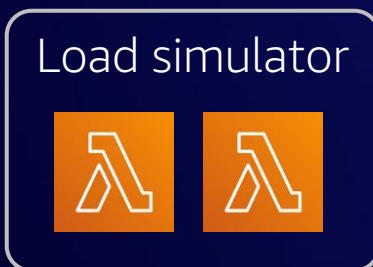
4

5

6

3

2



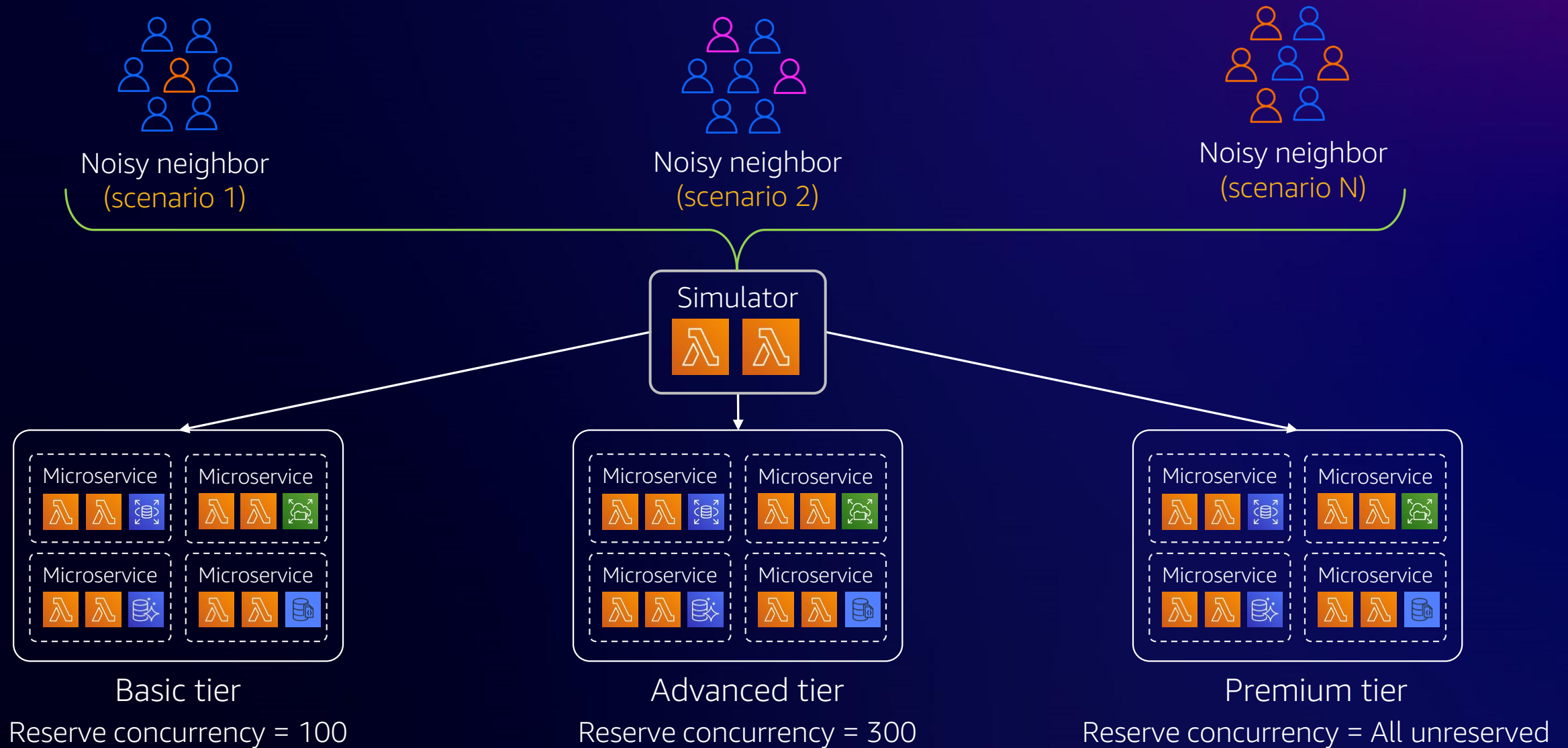
Async load
simulation

Provision
tenants

Application plane

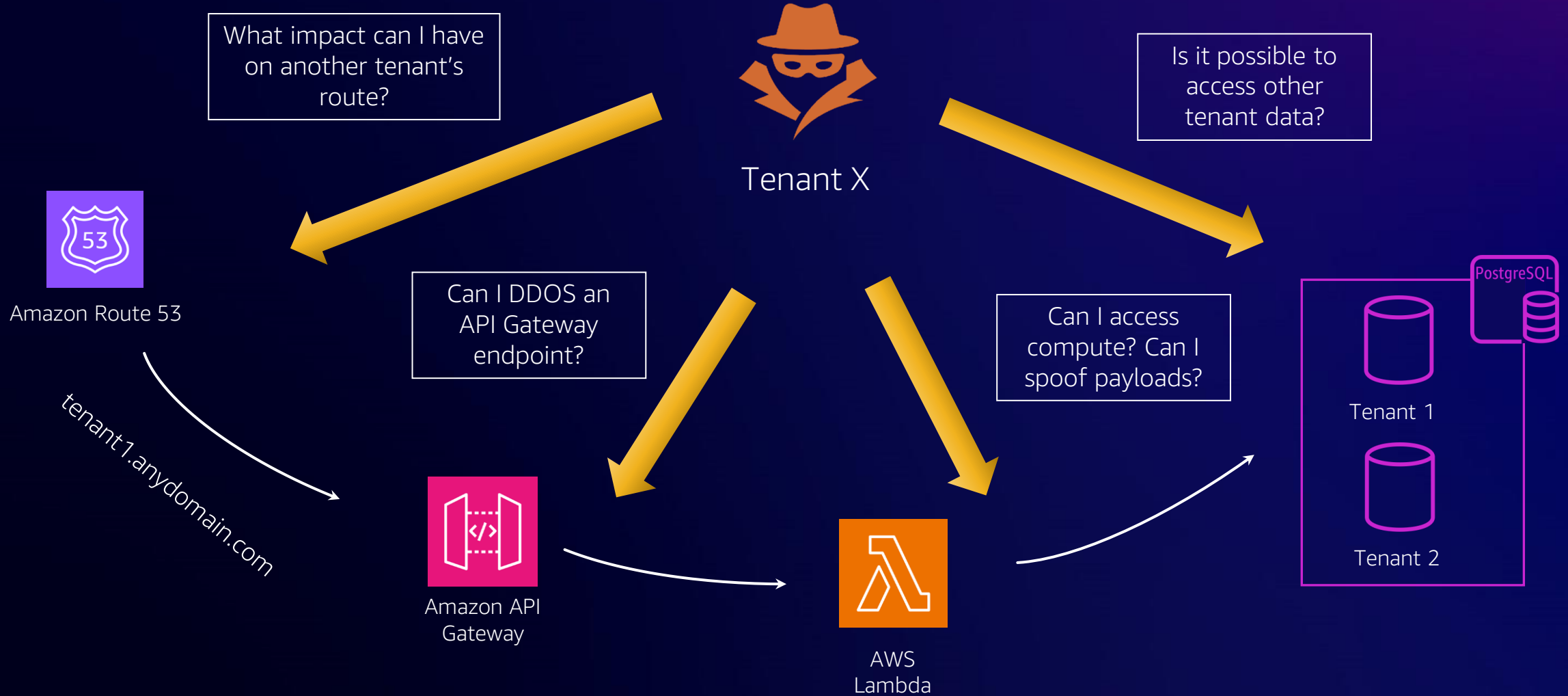
Control plane

Another approach: Provisioned concurrency

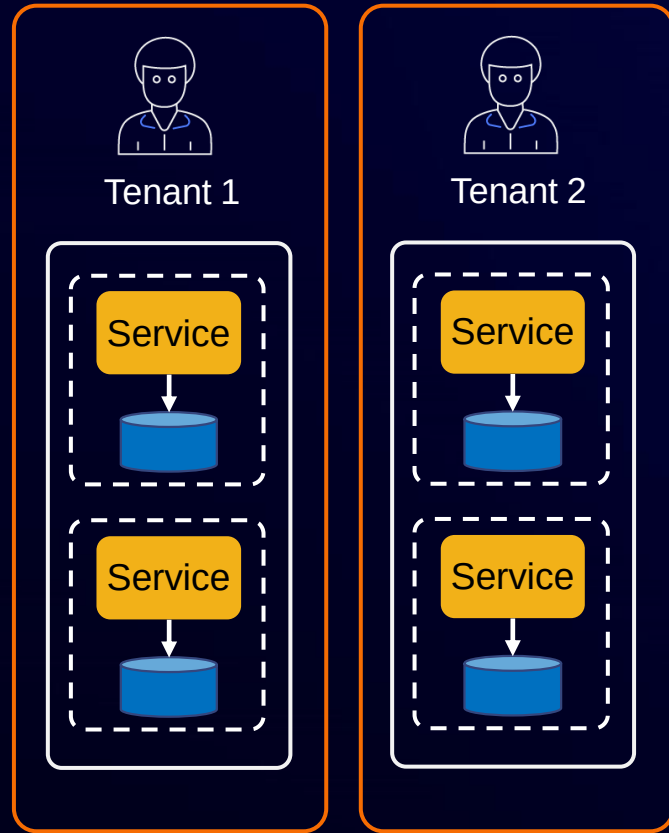


Chaos and tenant isolation

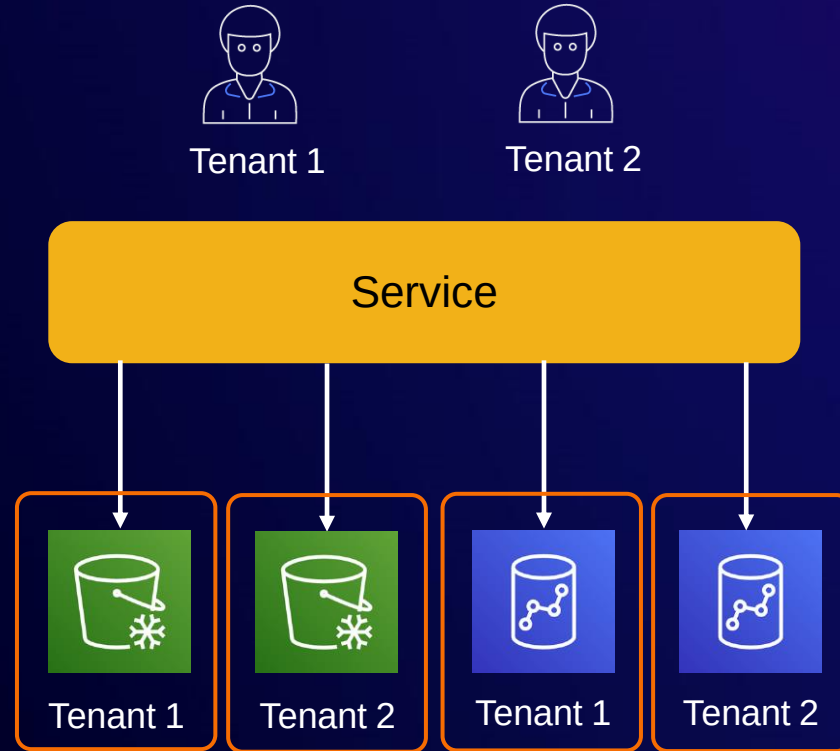
The challenge of tenant isolation



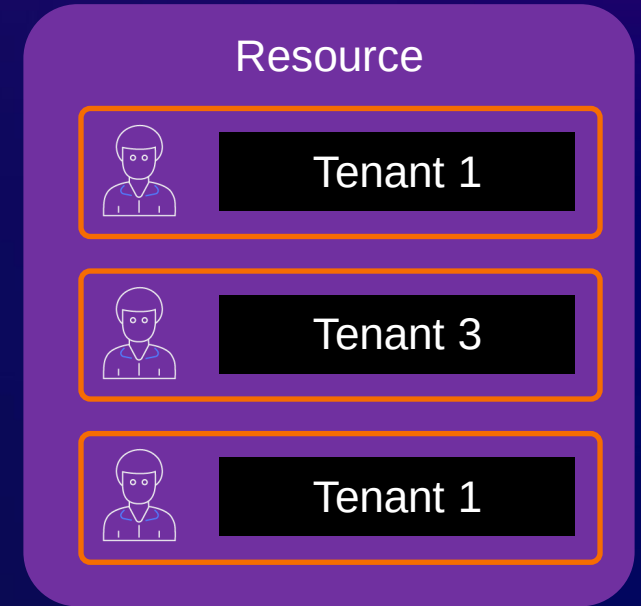
Multiple dimensions of isolation verification



Full stack
isolation

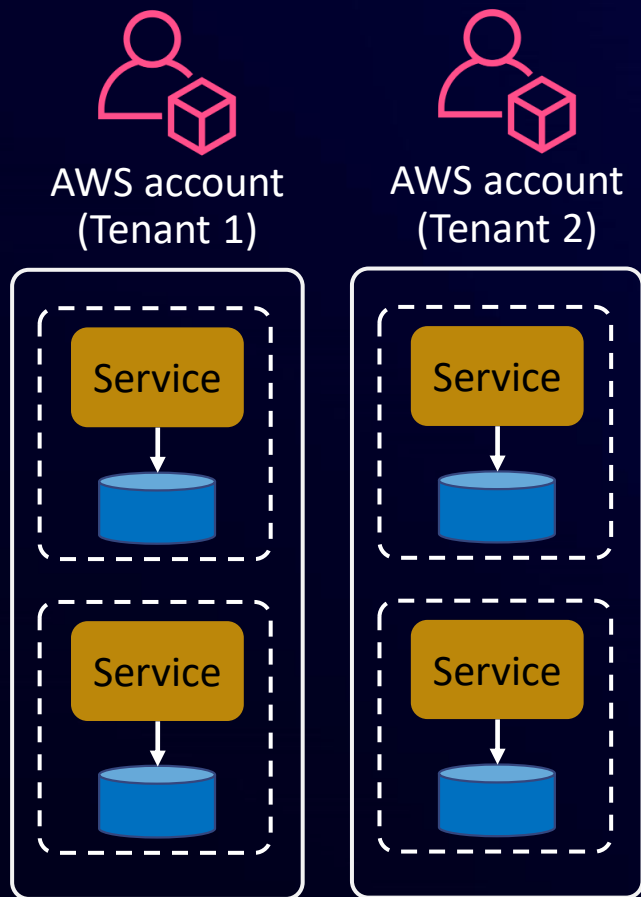


Resource-level isolation

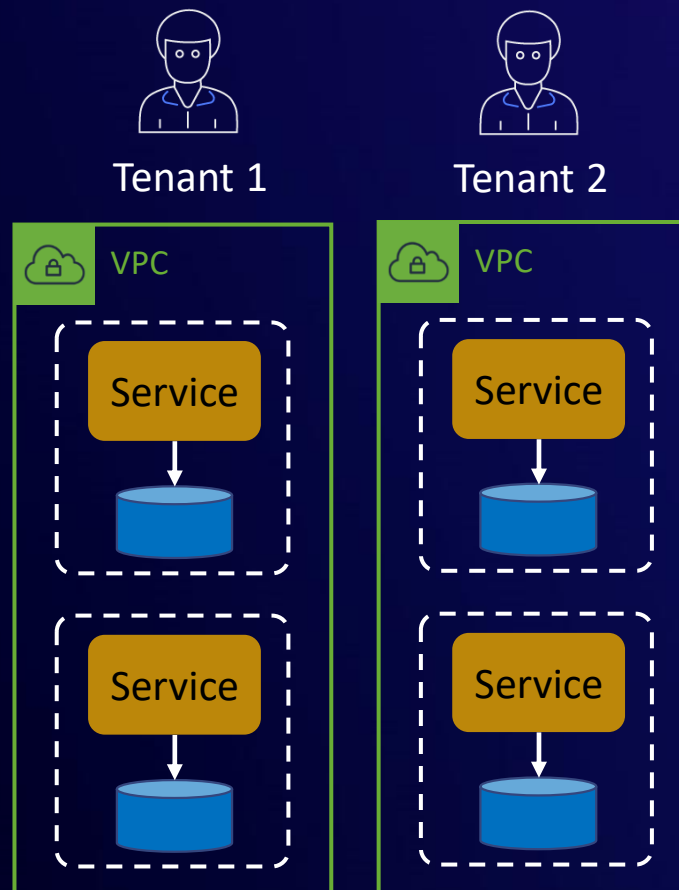


Item-level isolation

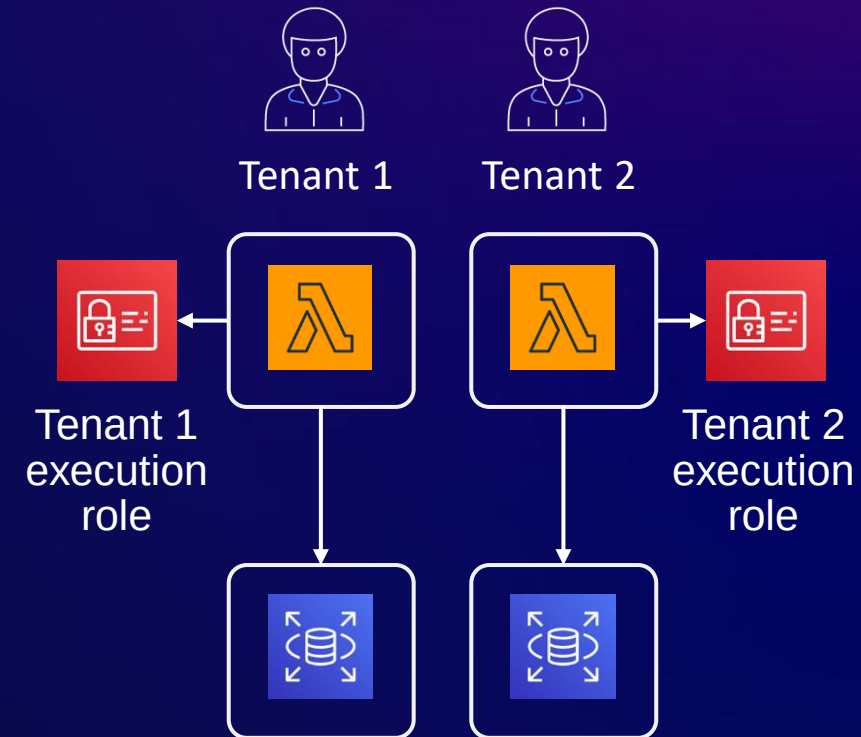
Testing silo isolation environments



Account per tenant



VPC per tenant



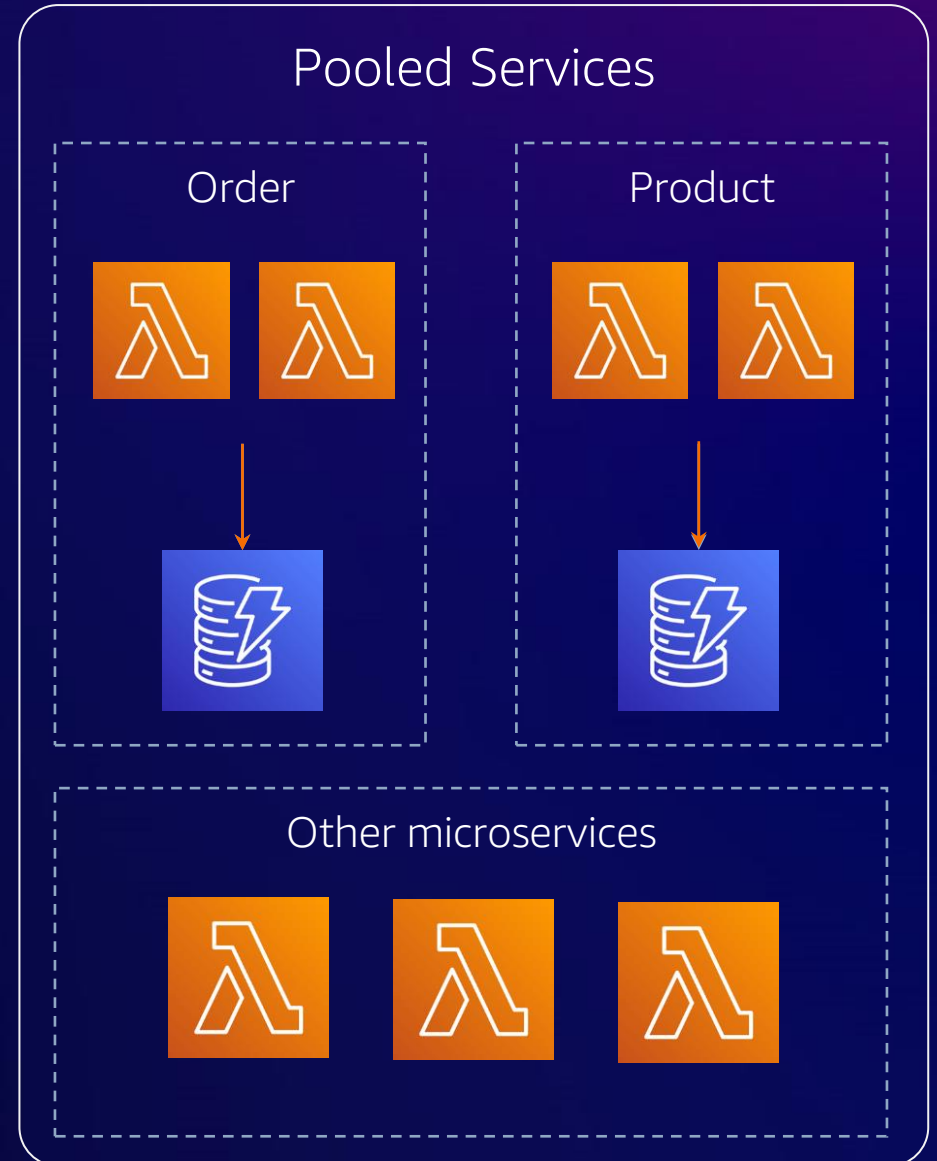
Silo with AWS Lambda

Testing tenant isolation with injection


Tenant 1


Tenant 2


Tenant 3



Testing tenant isolation with injection

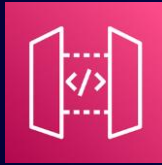
Tenant 1

Tenant 2

Tenant 3

JWT

Amazon API Gateway



AWS Lambda Authorizer

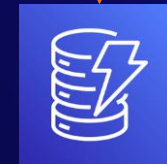


API keys & usage plans



Pooled Services

Order



Product



Other microservices



Testing tenant isolation with injection

Tenant 1

Tenant 2

Tenant 3

JWT

Amazon API Gateway



AWS Lambda Authorizer

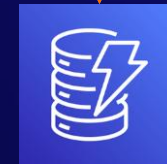


API keys & usage plans



Pooled Services

Order



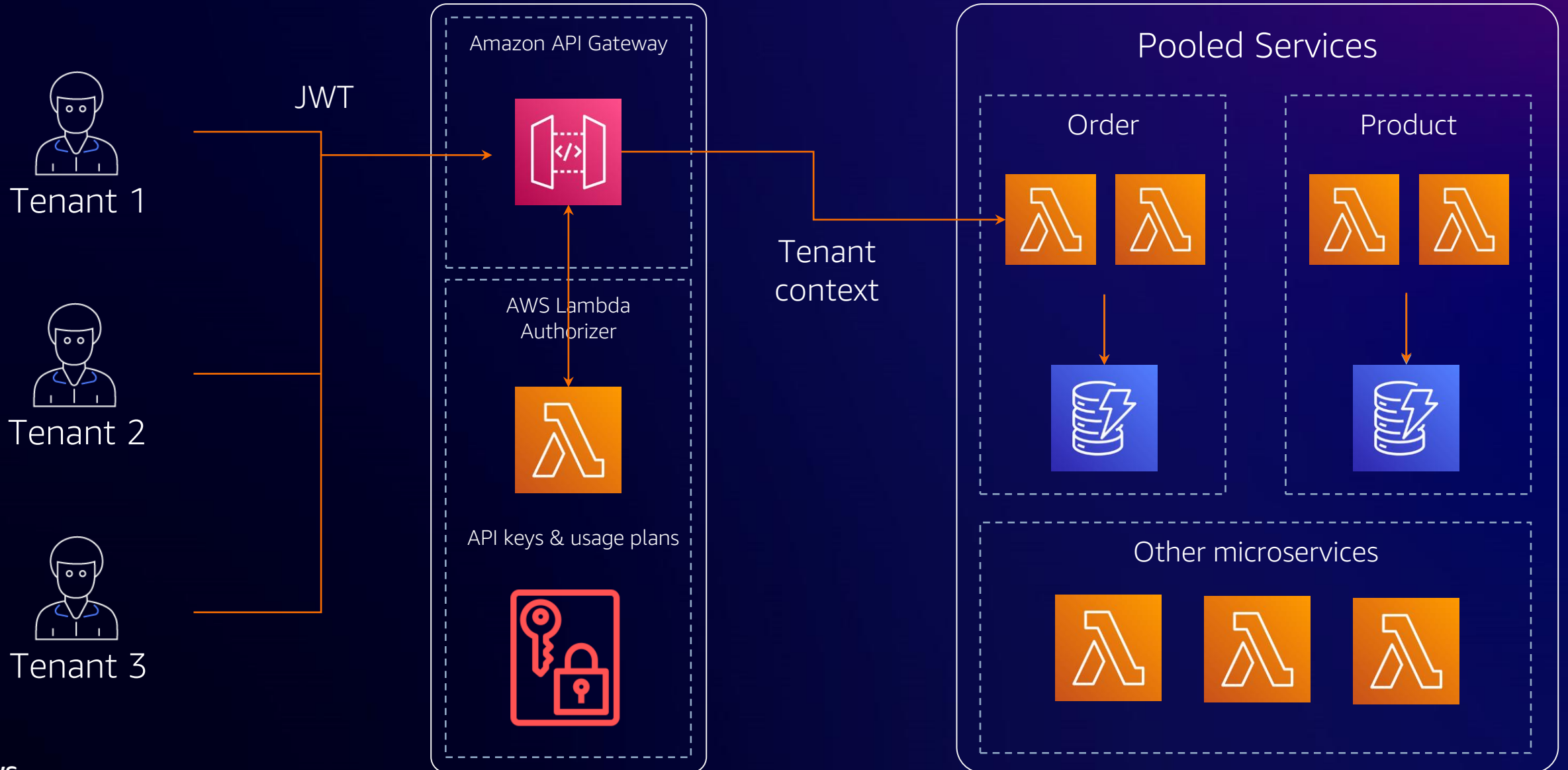
Product



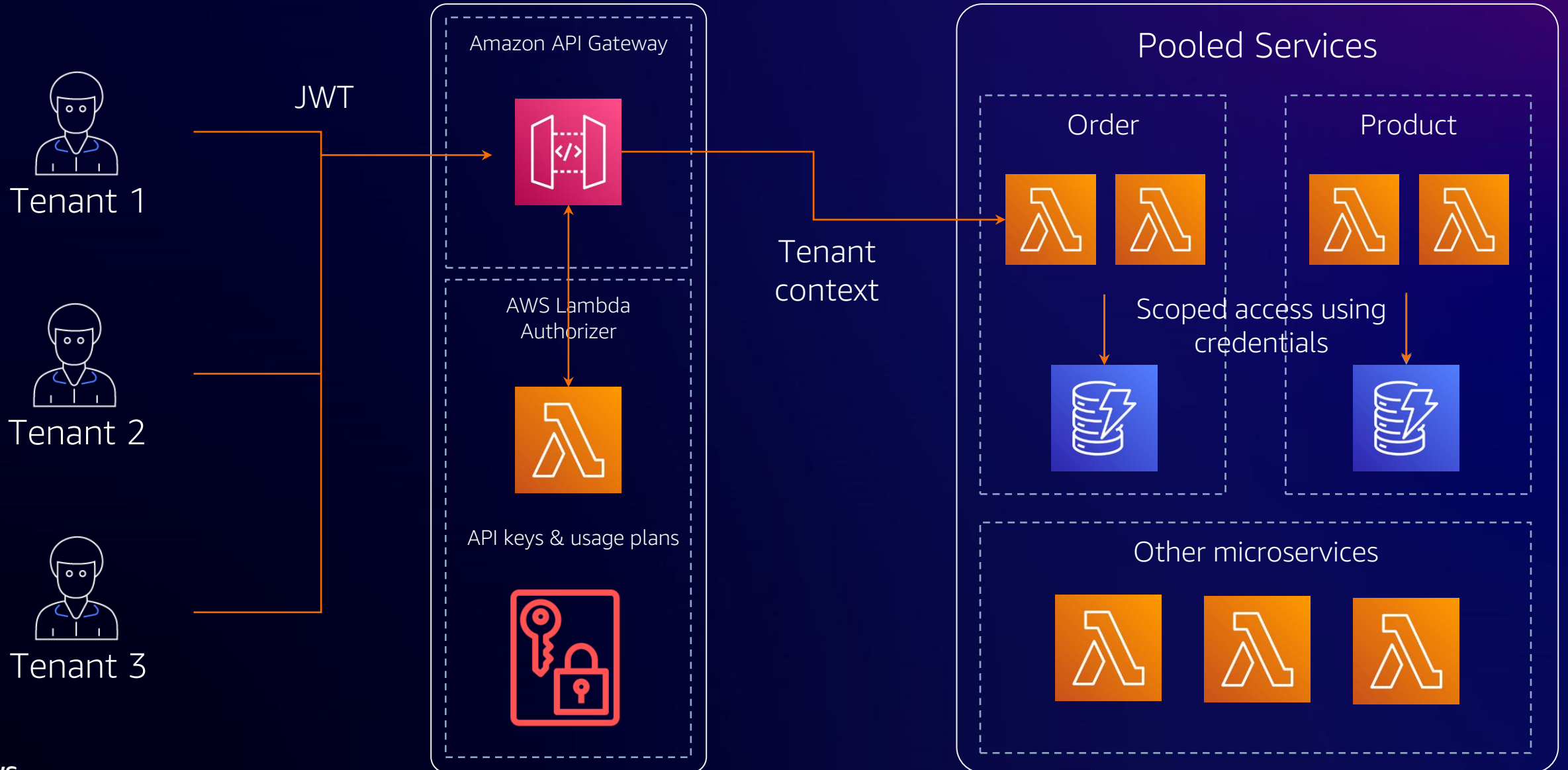
Other microservices



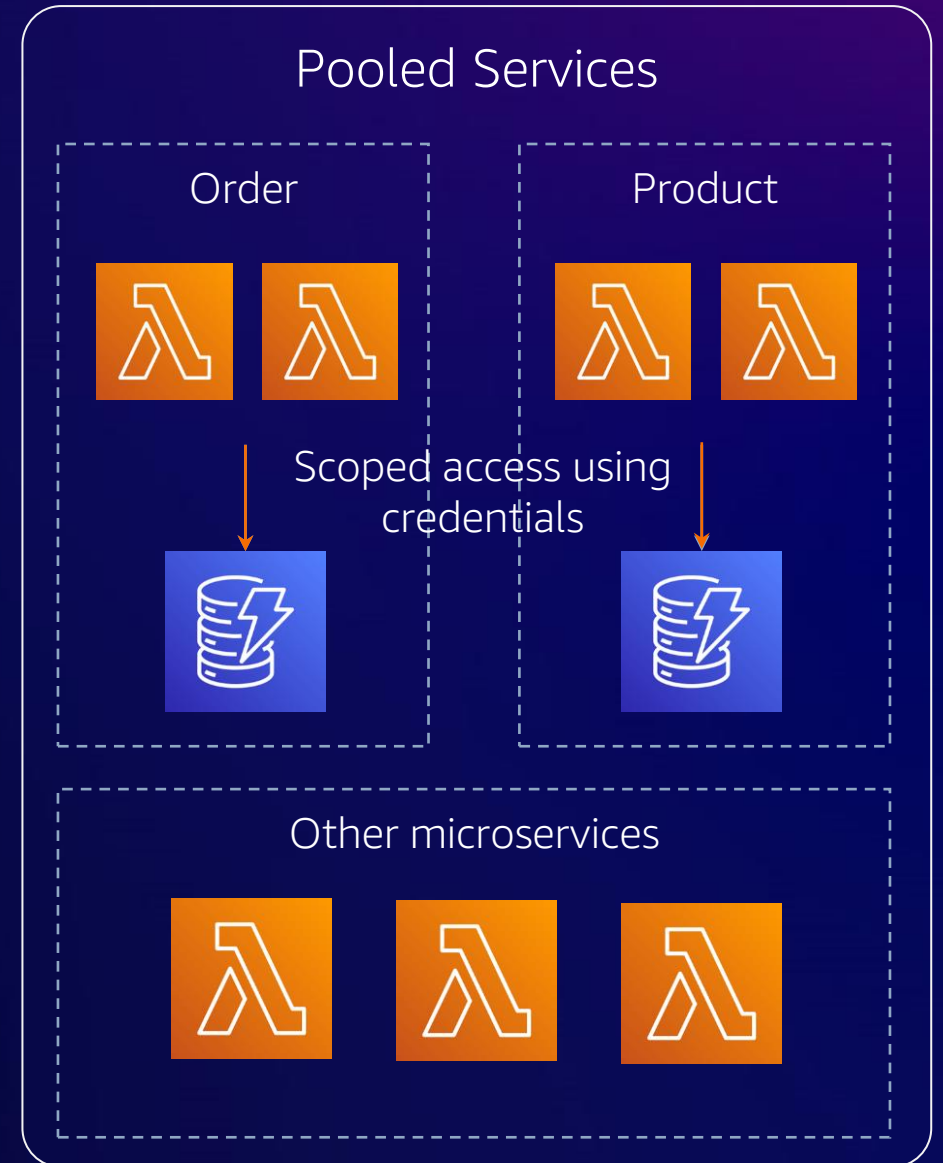
Testing tenant isolation with injection



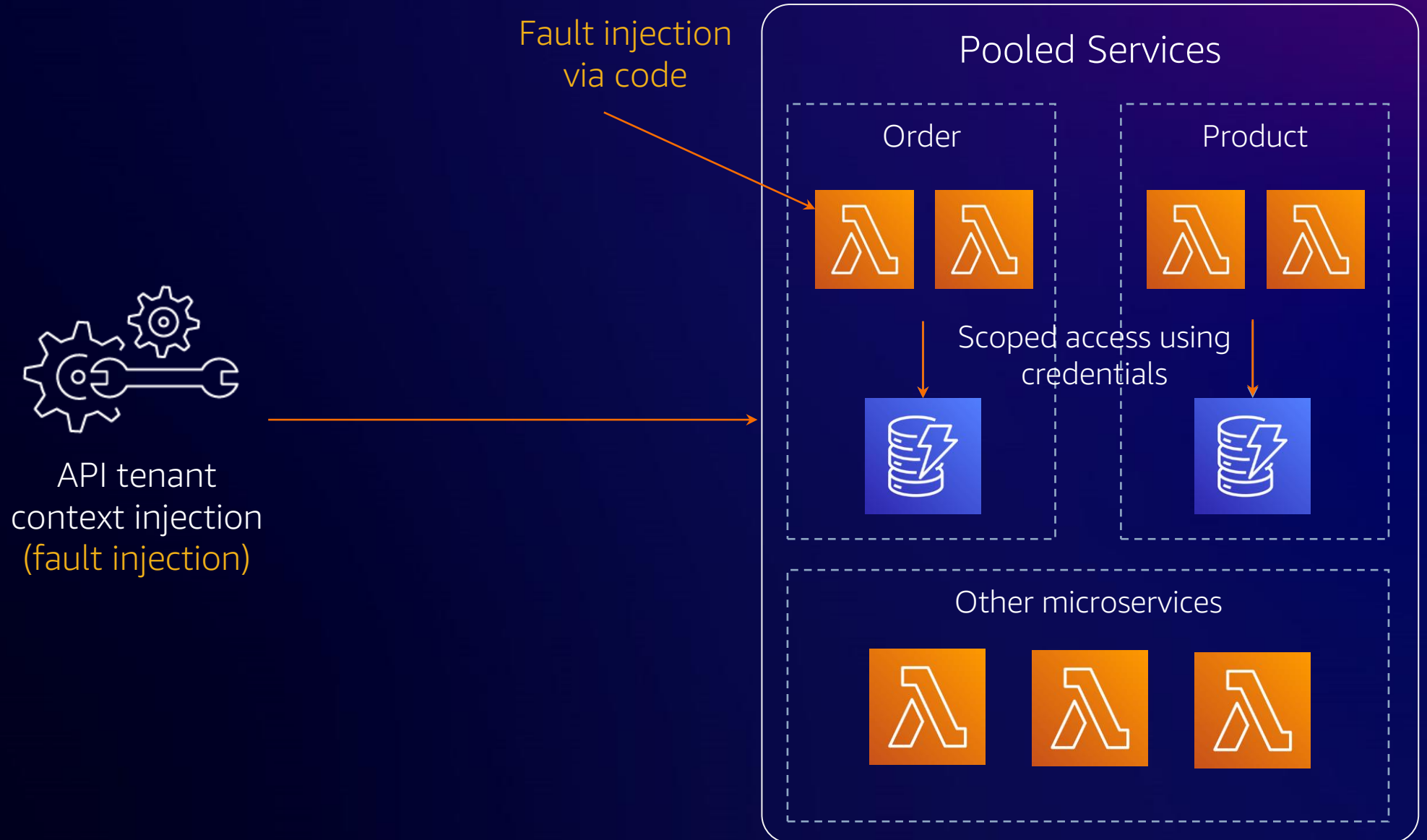
Testing tenant isolation with injection



Testing tenant isolation with injection

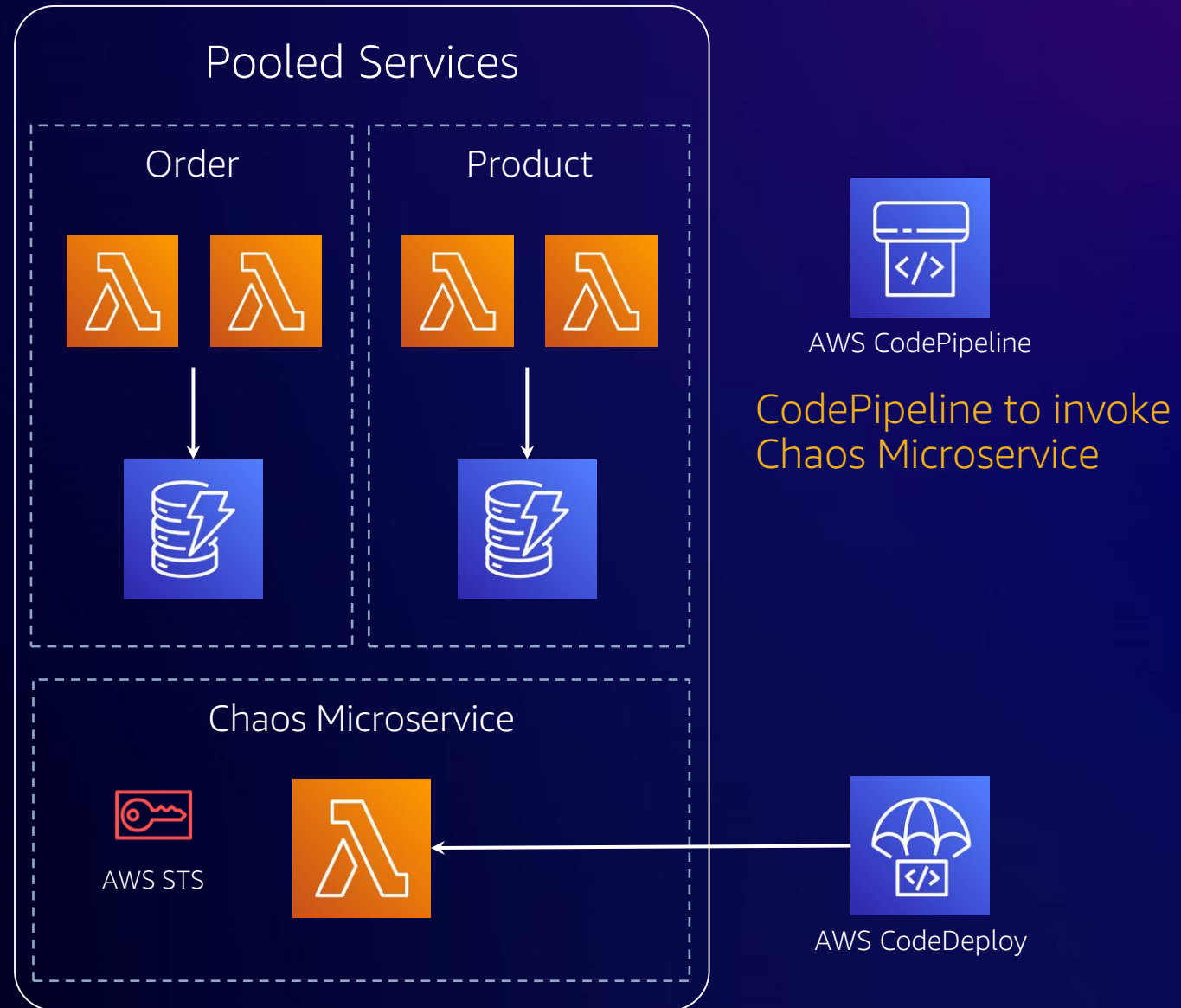


Testing tenant isolation with injection



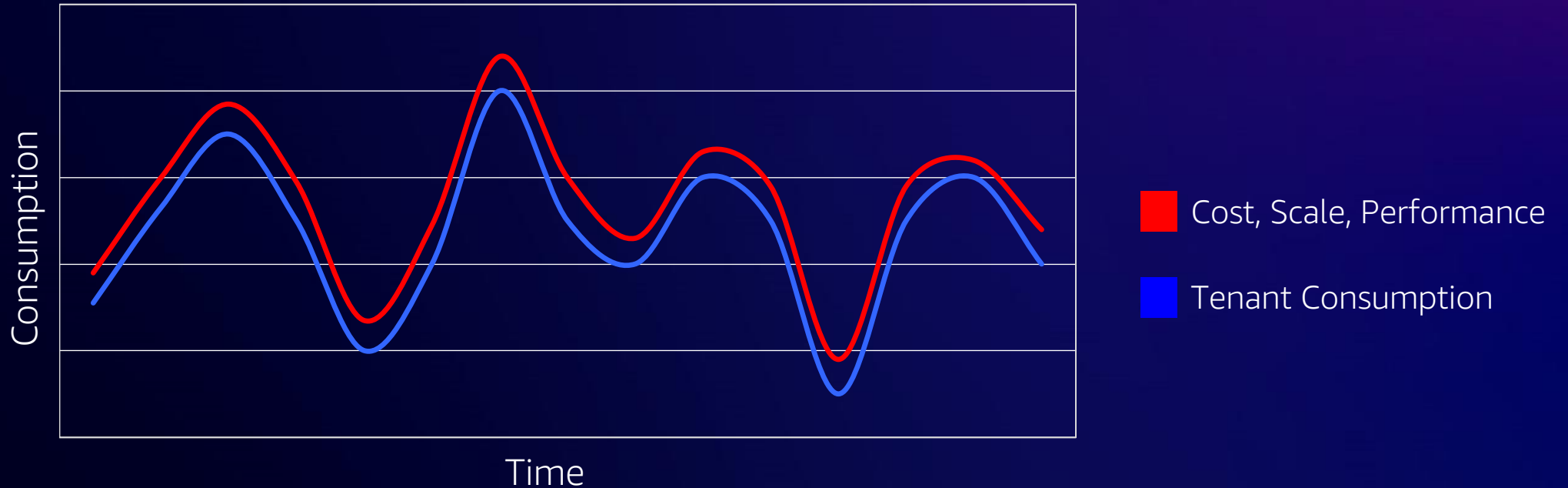
Another option

Use Assume Role (STS) to test dynamic policies for desired results



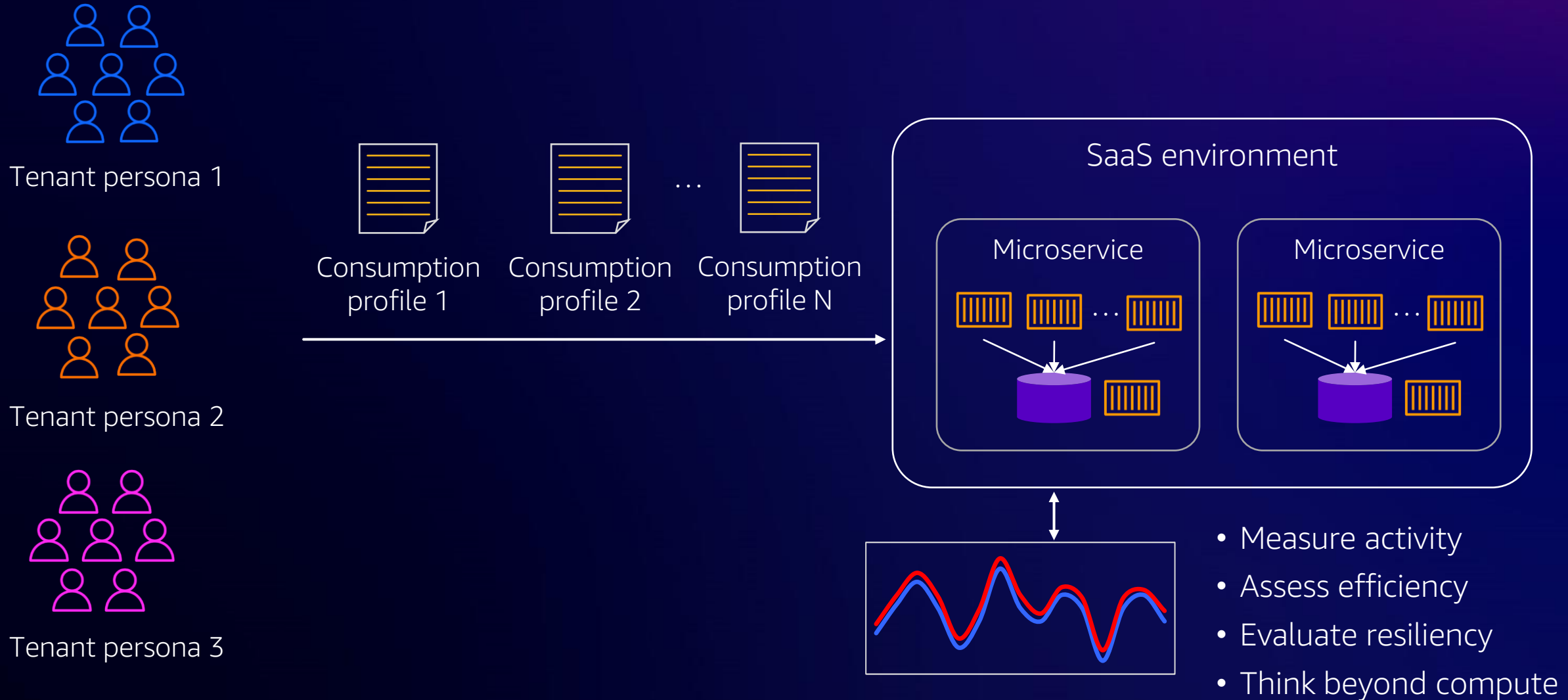
Chaos and multi-tenant scaling

How do you evaluate scaling?

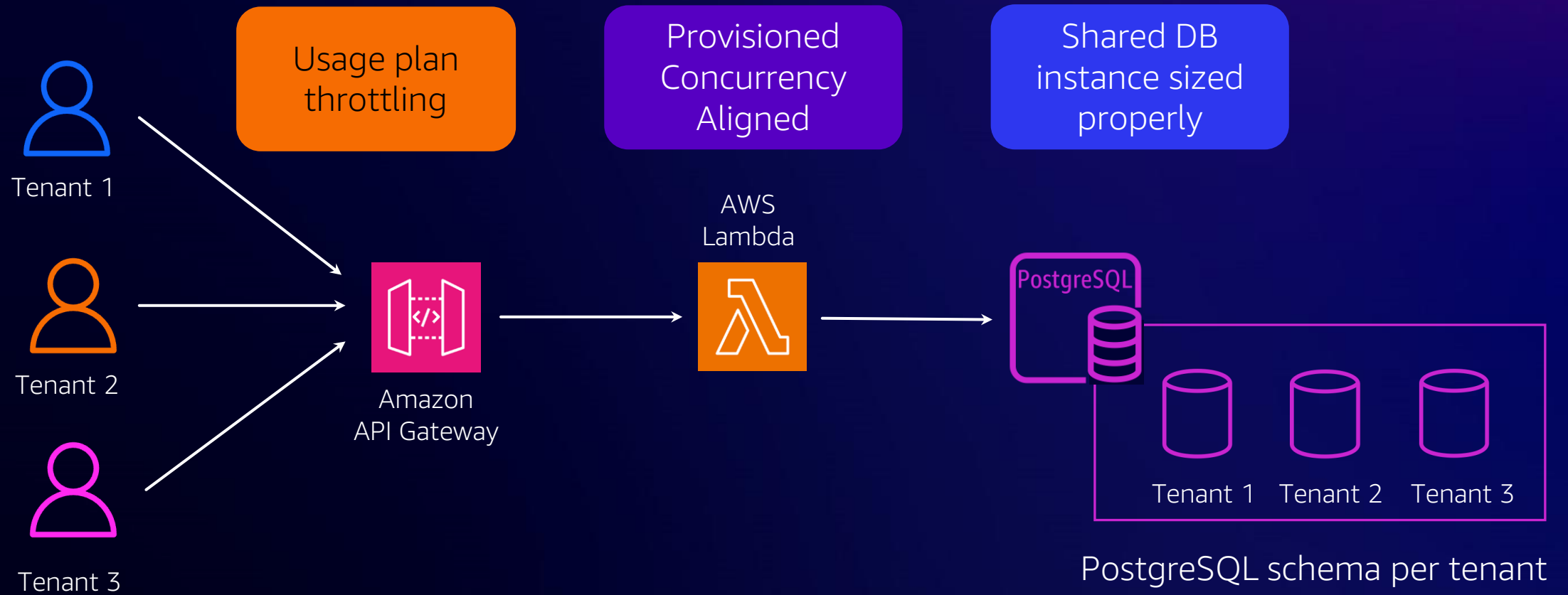


- Need to know if you're scaling based on your business and technical goals
- Assessing availability, resilience, and cost efficiency

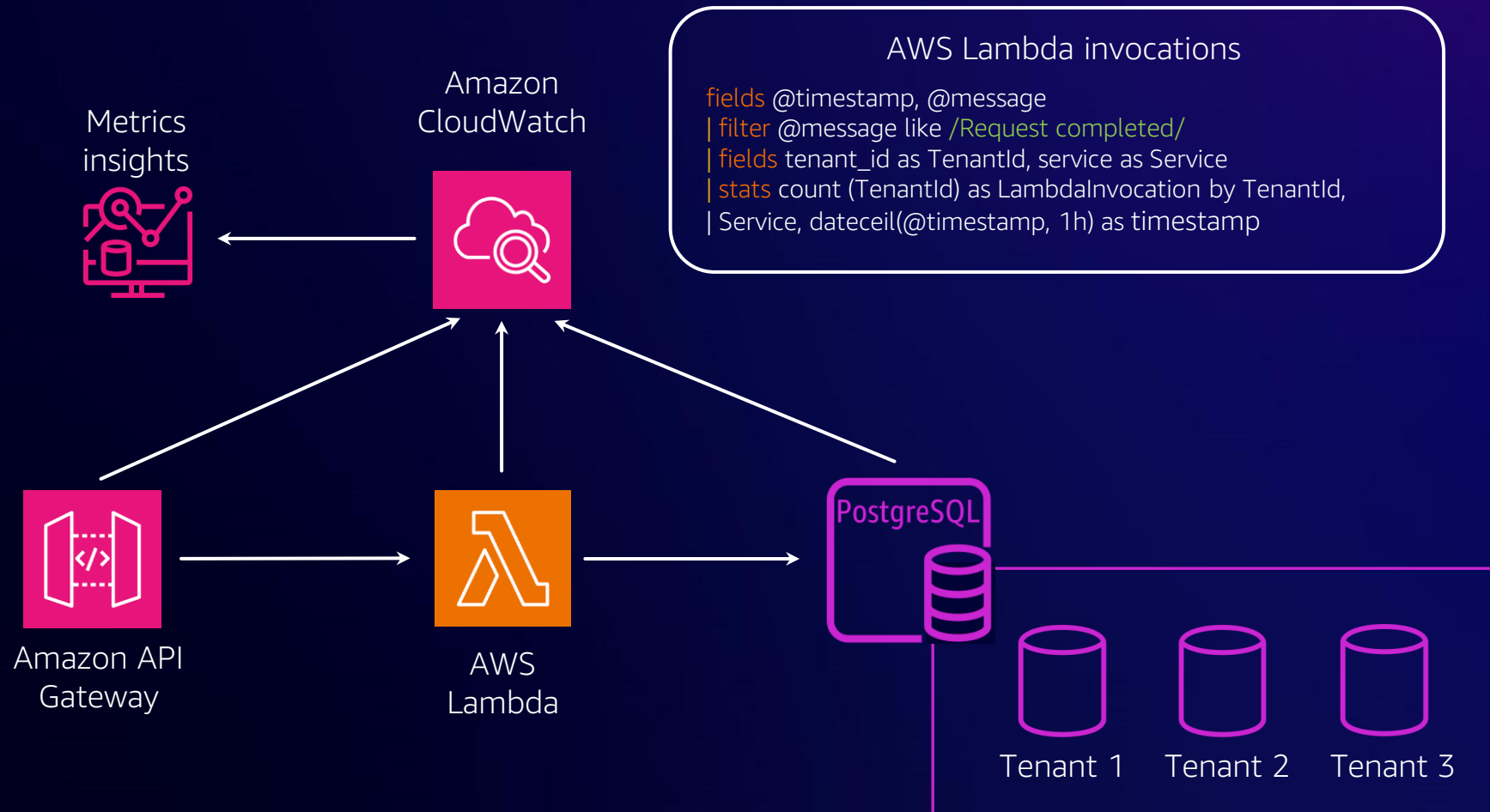
Defining your chaos scaling strategy



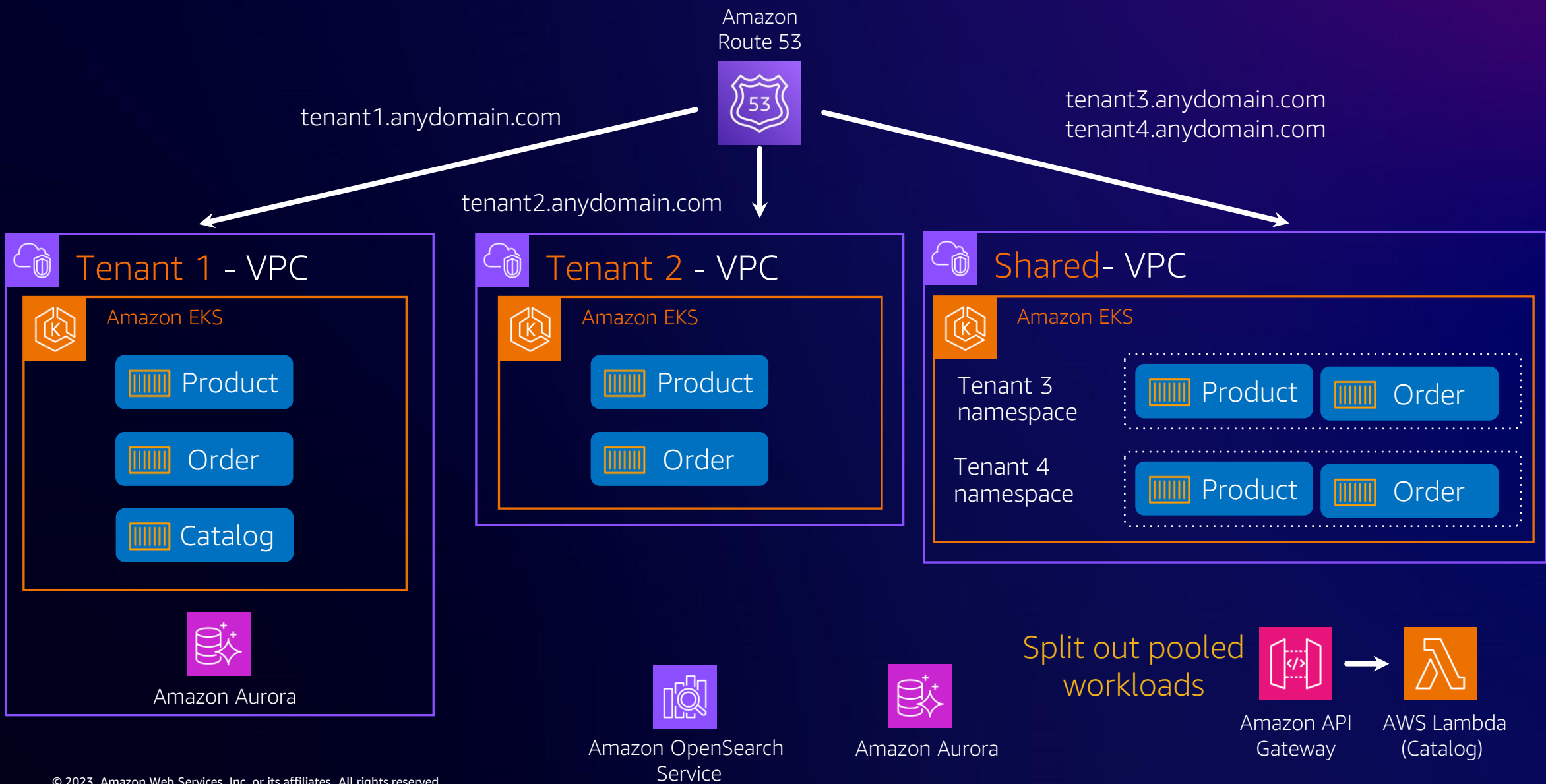
Serverless



Measurement of performance requires observation

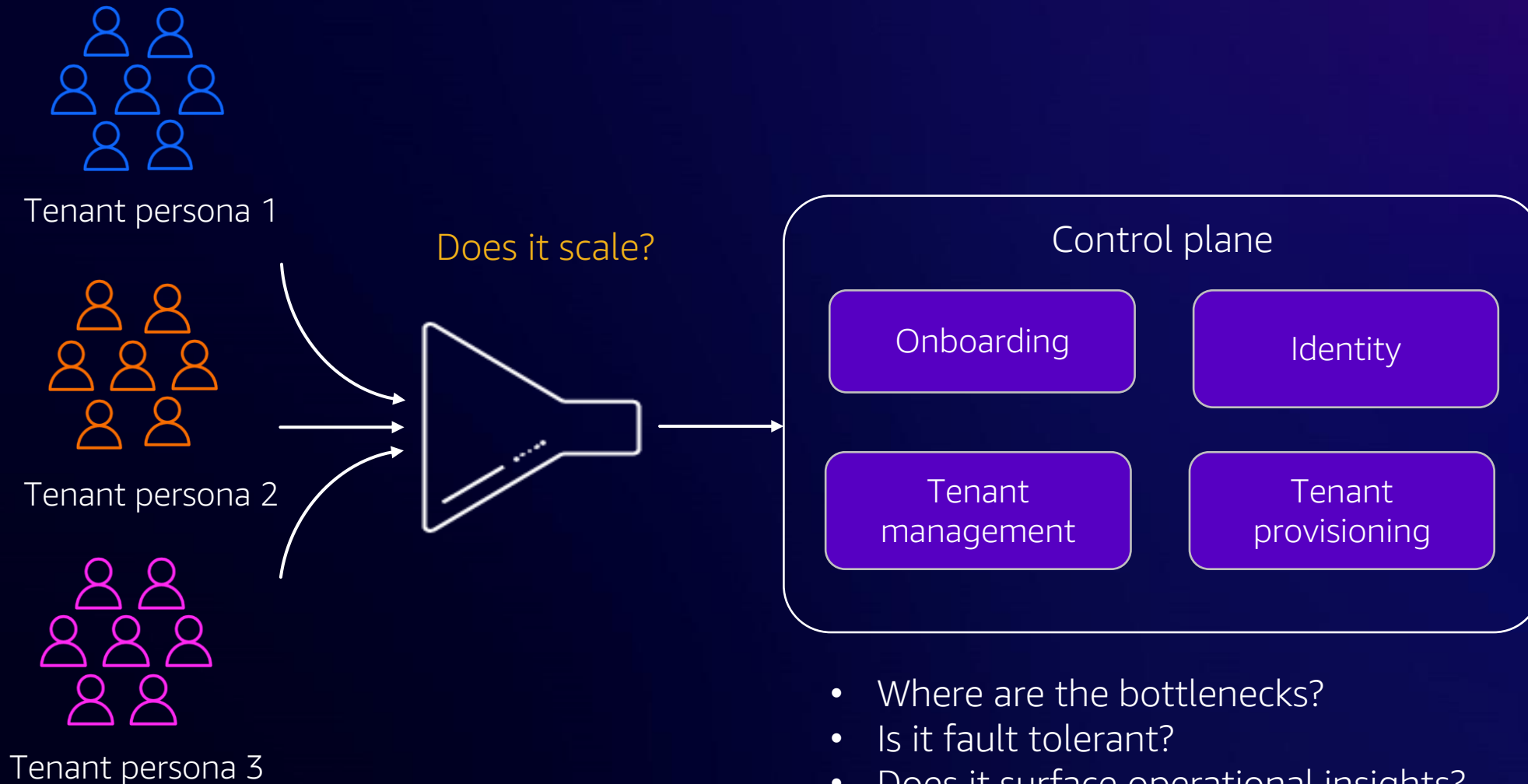


Considering how/where tiering impacts this



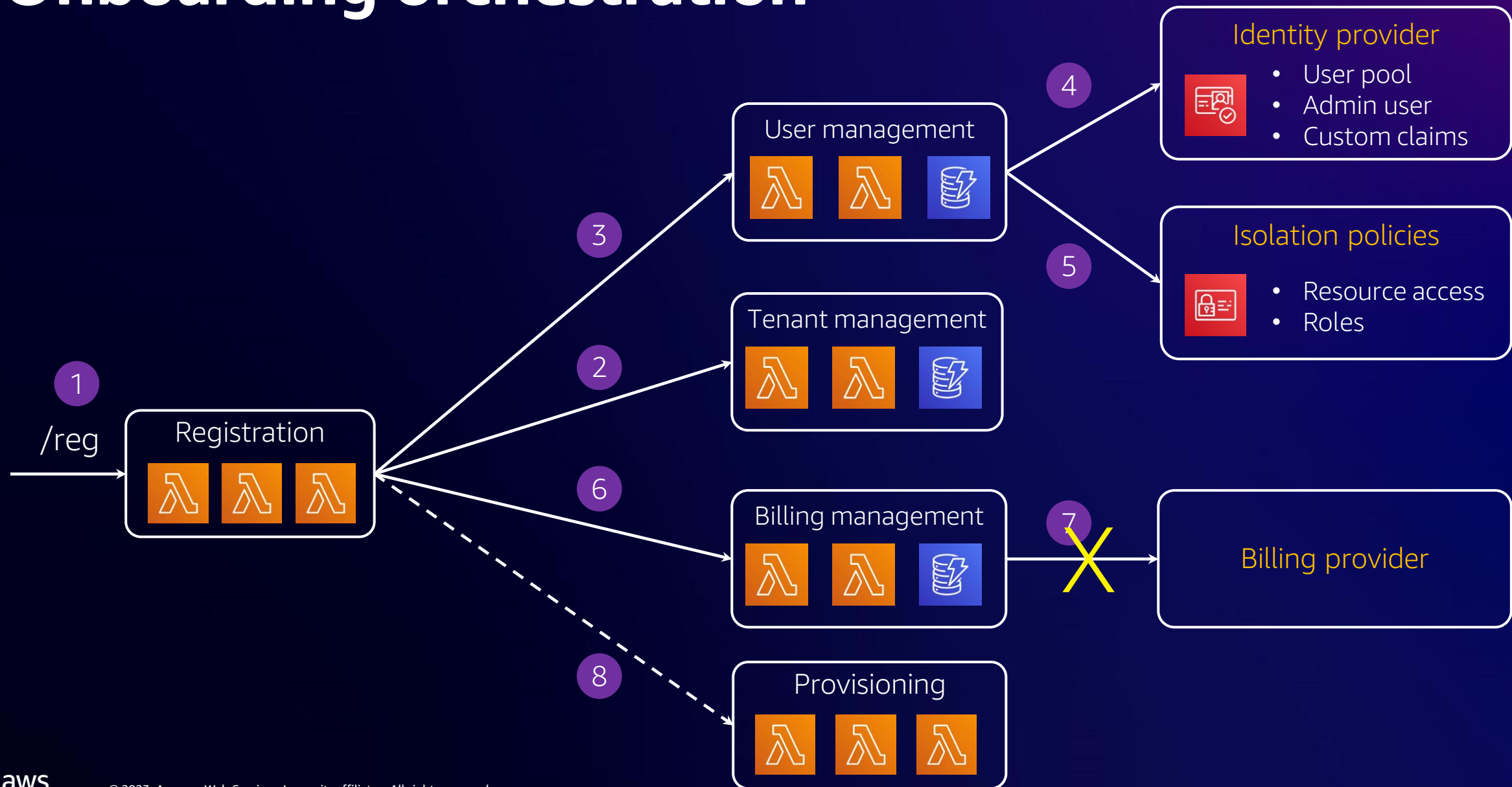
Chaos and onboarding tenants

Evaluating onboarding performance and resilience



- Where are the bottlenecks?
- Is it fault tolerant?
- Does it surface operational insights?
- Does it meet performance SLAs?

Onboarding orchestration



Global considerations



How will you
authenticate?



Do you have API support to
support your chaos
automation?



Can you streamline
onboarding for exercising
scale?



How will you manage the lifecycle
of these environments?



How/where are these tests
integrated into your development
process?

Final points

- SaaS testing/chaos takes effort and planning
 - Understand what is needed to create and tear down environments
 - This should test your onboarding experience across your tiers/SLAs
- SaaS adds complexity beyond normal testing
 - Noisy neighbor, tenant isolation, multi-tenant scaling and performance
- Chaos will happen, plan for it
- Make chaos happen and your systems will respond when the real thing does

Additional SaaS sessions

Breakout sessions

SAS305 – SaaS architecture pitfalls: Lessons from the field

SAS304 – SaaS deep dive: Inside a scalable, efficient multi-tenant architecture

SAS406 – SaaS DevOps deep dive: Automating multi-tenant deployments

SAS306 – SaaS meets AI/ML & generative AI: Multi-tenant patterns & strategies

SAS308 – SaaS anywhere: Designing distributed multi-tenant architectures

ARC311 – Building cost-optimized multi-tenant SaaS architectures

BWP301 – SaaS operations in action: Buy with Prime

CON313 – Deploying multi-tenant SaaS applications on Amazon ECS and AWS Fargate

Additional SaaS sessions

Chalk talks

SAS405-R – Federating identity in multi-tenant SaaS environments

SAS404 – Designing and implementing a SaaS cost-per-tenant strategy

SAS307 – SaaS tenant isolation: Patterns and strategies

SAS302-R – A look inside SaaS networking, routing, and integration patterns

SAS408 – Chaos and SaaS: Pushing your multi-tenant architecture

SVS320 – Building multi-tenant applications with AWS Lambda and AWS Fargate

ARC205 – SaaS jump-start: Exploring core multi-tenant strategies

BIZ207 – Integrate your SaaS application or security tool with AWS AppFabric

Workshops

SAS401-R – SaaS microservices deep dive: Multi-tenancy meets microservices

SAS402 – SaaS multi-tenancy meets Amazon EKS: A hands-on deep dive

SAS403 – SaaS survivor: Building a rich multi-tenant operations experience

SAS301 – SaaS and AI/ML: Inside a multi-tenant AI/ML solution

PEX401-R – Building a multi-tenant SaaS control plane: An inside look

Additional SaaS sessions

Builder session

SAS407 – Kubernetes DevOps tooling meets SaaS: Automating tenant provisioning

Business session

PEX205 – Benchmarking your SaaS business on AWS

Explore resources from AWS SaaS Factory



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Thank you!

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